Longchang Liu

Software Engineer

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Education

Hangzhou Dianzi University Aug 2022 - Jun 2026
B.S. in Computer Science and Technology

Experience

Siorigin - Agentic Infra (Kernel-Gen Agent) Jul 2026 - Present
An agentic infrastructure system for internal workloads: an agent autonomously generates, validates and tunes GPU kernels, and the system exposes a unified kernel generation and evaluation capability internally — reaching [N.Nx] geometric-mean speedup on FlashInfer benchmarks, on par with the published state of the art.

- **Generation**: led the end-to-end loop — the agent reads a target operator, writes candidate kernels, and iterates until a version passes correctness and beats the baseline.
- **Evaluation**: each candidate is compiled, checked against a PyTorch reference and profiled on the target GPU, then scored on measured latency and achieved bandwidth; both gates must pass before delivery.
- **Search and feedback**: profiler output is attributed to memory- or compute-bound and fed back as the next prompt, keeping only useful evidence and failure reasons to bound context growth, with iteration caps and regression rejection holding the search monotonic.
- **Delivery**: internal users only supply the target operator and a reference implementation to get back a validated, benchmarked kernel, with no GPU programming or tuning on their side.

Alibaba - Amap (Intern) Oct 2025 - Jan 2026
Responsible for productionizing the Order Analysis Assistant and AI Ride-Hailing LLM projects, improving complex task handling and integrating shared AI capabilities.

- On the Order Analysis Assistant, designed and launched Deep Research, improving complex task decomposition and handling stability in long-context scenarios.
- Participated in the cold start of AI Ride-Hailing, building unified and stable AI invocation interfaces to keep the core path reliably integrated.

Tencent - Ads ChatBI System (Intern) Apr 2025 - Sep 2025
Contributed to the NL2Chart core workflow of Tencent Ads ChatBI, enabling conversion from natural-language requests to charts.

- To guarantee source corpus quality, designed and implemented an automated corpus review system, raising core-workflow recall accuracy from 50% to 80%; also introduced a state machine for stability and crash recovery.

Open Source

LangChain4j (12k+ stars) contributor — Java ecosystem AI framework; community issue fixes and capability improvements.

Technical Skills

- **AI Infra**: Transformer architecture and KV Cache; agent context management, multi-agent collaboration patterns, harness design and eval systems, with hands-on GPU kernel generation and tuning.
- **CS Fundamentals**: Solid OS and networking fundamentals — memory management (page tables), system calls, thread scheduling, TCP/UDP and congestion control.
- **Java & Engineering**: JVM class loading, memory model and G1/ZGC; Synchronized, Volatile, AQS and thread-pool internals; SQL optimization, MySQL tuning and production troubleshooting.